

3 Exploring a Database in SQLite

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Exploring a Database in SQLite

Chapter Overview

- » Launching the SQL software ✓
- » Opening a database file ✓
- » The Database Structure tab ✓
- » The Browse Data tab ✓
- » The Execute SQL tab ✓
- » Viewing query results ✓
- » Data Analysis Checkpoint ✓

In this chapter, we will familiarize ourselves with the interface of our chosen SQL browser for SQLite: DB Browser.

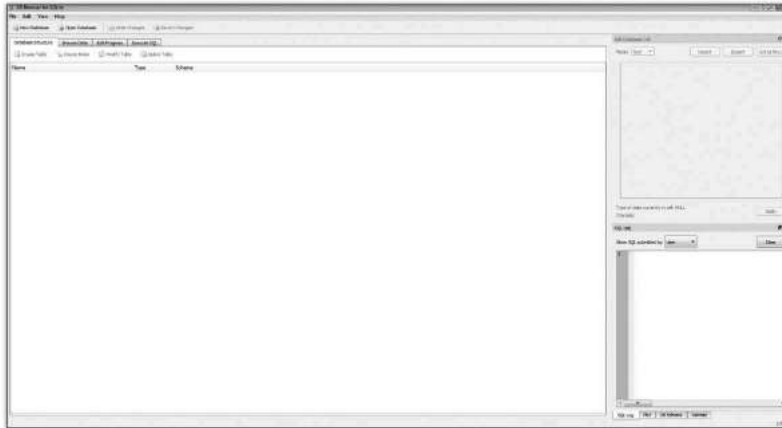
→ Environment Orientation

We have a bit more preparation to do and then we are ready to start analyzing the iTunes database. Now that we have installed DB Browser for SQLite software and downloaded the iTunes sample database, it is time to fire it up and get going!

1. Start the DB Browser for SQLite application.
 - a. **Mac users:** Go to Finder and double-click DB Browser for SQLite in your Applications folder.

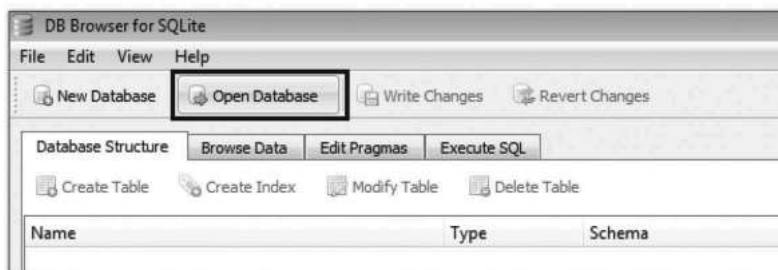
- b. **Windows users:** Go to your Start Menu and click on the DB Browser for SQLite application from your list of installed programs.

2. You will see the following default screen (Figure 23):



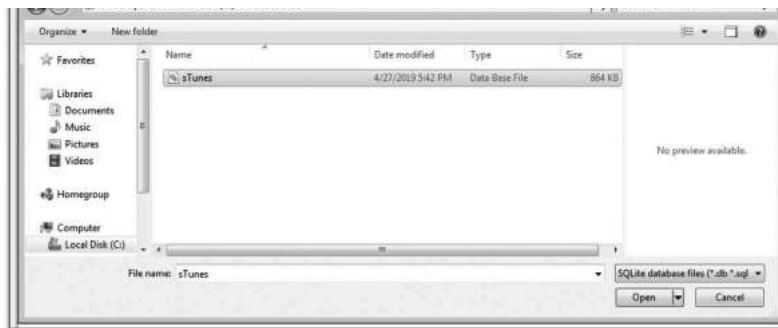
Opening the sTunes Database

3. With DB Browser open, click on "Open Database."



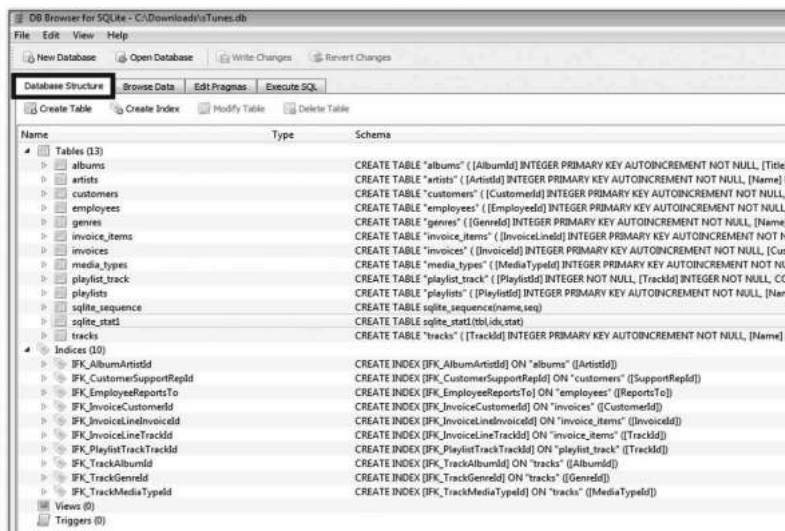
4. This opens the "Choose a Database File" dialog box. Navigate to the folder where you downloaded the sTunes sample database file and click on "Open."





Investigating the Structure of the Database

When the file is opened, the Database Structure tab shows the tables contained in the sample database.



Now that we can investigate the database structure, we can learn a lot more about our fictitious company, iTunes. Once the IT department gives us the proper permissions, our company will most likely give us some time to familiarize ourselves with the database we were tasked to analyze. Before we start writing any SQL statements, it's a good idea to see what type of data we have available to us.

First, we can see that there are thirteen tables in the iTunes database. By expanding the arrow to the left of the table name, we are shown the fields contained in each table. In the *albums* table, as can be seen in Figure 27, the fields are AlbumId, Title, and ArtistId.

IMAGE fig. 27

Database Structure		Browse Data	Edit Pragmas	Execute SQL
Create Table		Create Index	Modify Table	Delete Table
Name	Type			
Tables (13)				
albums				
AlbumId	INTEGER			
Title	NVARCHAR (160)			
ArtistId	INTEGER			

REMEMBER

At the beginning of chapter 1, we talked about the fundamental terminology of this book. The fields we are seeing are the columns in each table. The data stretched across the rows of the table (not pictured here) are the records.

The AlbumId field is an *integer* data type, which means it holds numeric data (numbers).

The Title field is a character data type (known as *NVARCHAR*), which means it holds characters or non-numeric data.

The ArtistId field is also an integer data type.

REMEMBER

We spoke about metadata and different data types in chapter 1, where we went over database structure. The Type field in our database browser panel (see Figure 27) is a good example of metadata. The data type for each field is chosen based on the most logical type for its function.

Viewing the Individual Records

The Browse Data tab can be used to see the records contained in each table. Use the drop-down to switch between tables and browse the data contained in each one.



AlbumId	Artist	Title	Year
1	For Those About To Rock We Salute You		1
2	Balls to the Wall		2
3	Rainbow and Wild		3
4	Let There Be Rock		4
5	Big Ones		5
6	Jagged Little Pill		6
7	Pearl Jam		7
8	Warner 25 Years		8
9	Playa Matador by Pearl Jam		9
10	Audioslave		10
11	Out of Exile		11
12	Backstreet Backtrack		12
13	The Best of Billy Cobham		13
14	Absolute Punked: Friendship Love! (Disc 1)		14
15	Absolute Punked: Friendship Love! (Disc 2)		15
16	Black Sabbath		16
17	Black Sabbath Vol. 4 (Remastered)		17
18	Body Count		18
19	Chemical Wedding		19
20	The Best of Buddy Guy - The Millennium Collection		20
21	Friends Reunite		21
22	Quelque chose au Vieux		22
23	Mama Mia! Live		23

In Figure 28, the Browse Data tab displays the data stored in the albums table. It shows the columns we observed in the Database Structure tab. It also shows the actual data contained in these columns.

In row 1, the following data is contained in each column of the albums table:

AlbumId = 1

Title = "For Those About to Rock We Salute You"
ArtistId = 1



REMEMBER

As per our chapter 1 terminology discussion, the columns you are seeing are referred to as fields. Every row represents an individual record.

By examining this table, we can also learn a bit about the other tables in this database. In chapter 1, we introduced the concept of a foreign key field. In this table we can see that the primary key is AlbumId (there is a unique number for each record). We also have an ArtistId which accepts an integer instead of an actual artist name. This gives us a hint that there is probably another table that holds the actual artist name as a character data type and that ArtistId is most likely a foreign key field.



REMEMBER

A foreign key is a field that exists as a primary key in another table. If we look at the artists table, we can confirm that ArtistId is the primary key in that table, so it exists as a foreign key in this one.

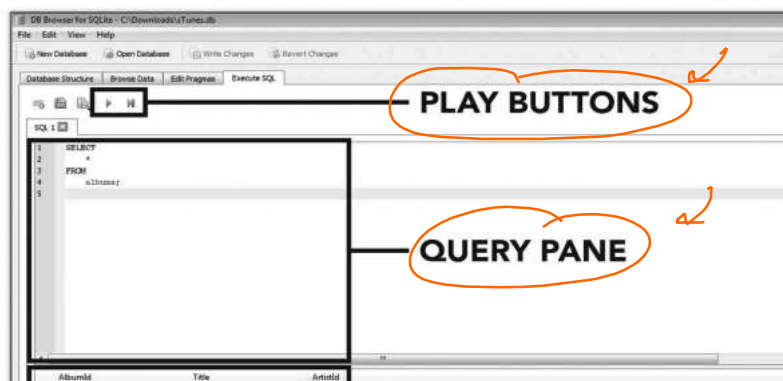
The Execute SQL Tab

The Execute SQL tab is where you write your SQL statements. There are three main component window panes in the execute SQL tab: the Query Pane, the Results Pane, and the Messages Pane. Let's explore these three panes by writing the following SQL code into the Query Pane, as seen in Figure 29:


```
SELECT
 *
FROM
 albums;
```

This SQL statement selects all fields (the * symbol designates "all fields") from the *albums* table and, once you press the play button above the Query Pane, displays the fields, and data in those fields, in the Results Pane below.

As seen in Figure 29, the Play button to the left executes all SQL statements written in the window. The Play button to the right executes only the SQL statement that your cursor is on (it executes only one statement at a time).



1	1	For Those About To Rock We Salute You	1
2	2	Balls to the Wall	2
3	3	Restless and Wild	2
4	4	Let There Be Rock	1
5	5	Big Ones	3
6	6	Jagged Little Pill	4
7	7	Facelift	5
8	8	Werner 25 Anos	6
9	9	Plays Metalica By Four Cellos	7
10	10	Audioclave	8

147 rows returned in 0.00 sec from: SELECT * FROM albums;

RESULTS PANE

MESSAGES PANE

NOTE

Since we are starting off with basic queries, we are only going to be running one SQL statement at a time. More advanced queries can use multiple statements executed at the same time. The ability to execute statements individually is valuable in multi-statement queries. If you

have only one complete SQL statement in the Execute Pane, then these Play buttons essentially do the same thing.

The Results Pane lists the output generated by our query. If we have more results than can be viewed on the screen at one time (Figure 30), the browser provides us with a scroll bar to navigate down through all the results.

There is another way to determine if what we see on the screen (in this case only five results), represents the entire query or if there is more data available. Under the Results Pane is the Messages Pane, which displays informative messages about our query.

- » The number of rows returned from our SQL statement
- » How much time our query or SQL statement took to run
- » Error messages, if our SQL statement contained errors

IMAGE fig. 30

	AlbumId	Title	ArtistId
1	1	For Those About To Rock We Salute You	1
2	2	Balls to the Wall	2
3	3	Restless and Wild	2
4	4	Let There Be Rock	1
5	5	Big Ones	3

 IMAGE fig. 31

```
347 rows returned in 9ms from: SELECT
*
FROM
    albums;
```

The first thing we see is that our query returned 347 rows of data. If we did not notice the scroll bar, then this message would serve as our notification to scroll down. The second thing we observe is that our query was returned in 9 milliseconds (abbreviated as 9ms). You may wonder why it is valuable to see how much time it takes to process a query. In this case, it doesn't really tell us anything. However, in larger databases with more advanced queries, it can take much longer to return data. The length of time it takes also depends on how a database is optimized and structured. We will talk more about database structure in chapter 6 when working with multiple tables.

 NOTE

The Messages Pane is also the location where any error messages will be displayed. If your query does not execute correctly, check the Messages Pane.

Data Analysis Checkpoint

 NOTE

You have reached the first Data Analysis Checkpoint in this book!
These checkpoints will be at the end of every remaining chapter. This
is a good time to practice what you have learned so far.

Using the Database Structure tab and the Browse Data tab, try to answer
the following questions:

1. How many tables are in our database?
2. How many fields does the table named *tracks* have?
3. What are some of the data types in this table?
4. What does the actual data look like in the table?

 REMEMBER

The answers to these Data Analysis Checkpoints are at the end of the
book in Appendix I.

Chapter Recap

» DB Browser for SQLite, the chosen software for this text, opens like any other computer software program.

» Selecting the Open Database button will allow you to select a database file from your computer to open.

» Investigate the structure of the database file with the Database Structure tab.

» Use the Browse Data tab to view individual records in the table you select with the provided drop-down menu.

» Use the Execute SQL tab to write and then execute your first SQL statement.

» The Results Pane contains the data returned by your query.

» The Messages Pane contains information about your query including how many rows were returned, how long the query took to run, and any applicable error messages.

